Process Characterization Report

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCR-005

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

Table 2: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Revision history

Version	Effective date	Description of change
1.0	2026-07-24	Initial issue
2.0	2026-09-30	Re-analysis under INV-001: a-priori intermediate acceptance criteria, fragments by nrCE-SDS, two-sided capability for acidic variants, execution dates recorded (see INV-001)

Abbreviations

AMD, analytical method decision (AMD-001 set nrCE-SDS as the method for fragments); AMV, analytical method validation; BLA, Biologics License Application; CI, confidence interval; CPP, critical process parameter; CQA, critical quality attribute; CV, column volume (as a unit) or coefficient of variation (as %CV); DNA, deoxyribonucleic acid; ELISA, enzyme-linked immunosorbent assay; Fab, antigen-binding

fragment; Fc, crystallizable fragment; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight species; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; LMW, low molecular weight species; MSAT, Manufacturing Science and Technology; NOR, normal operating range; nrCE-SDS, non-reduced capillary electrophoresis with sodium dodecyl sulfate; PAR, proven acceptable range; PRESS, prediction error sum of squares; qPCR, quantitative polymerase chain reaction; QA, Quality Assurance; RSM, response surface methodology; SEC, size exclusion chromatography; SOP, standard operating procedure; WC-CPP, well-controlled critical process parameter.

Executive summary

This report presents the process characterization of the Protein A chromatography step (Step 5) of the A-Mab drug substance process. Protein A chromatography is the capture step of the purification train. It removes most of the host cell protein and DNA of the clarified harvest, and it adds one impurity of its own, Protein A ligand that leaches from the resin. The studies were run under the approved plan PCP-005 in a scale-down model of the commercial column, between 2025-11-05 and 2026-09-11. The report follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), the enhanced approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012) and the A-Mab case study (CMC Biotech Working Group 2009). This is Version 2.0 of the report. It re-analyses the same experiments as Version 1.0 against the criteria of PCMP-001 Version 2.0, issued under INV-001, and adds the fragment results by nrCE-SDS.

The step has 6 process parameters. Of these, 4 were studied in a 19-run screening design followed by a 28-run response-surface design, and 2 were studied one factor at a time. Protein load and elution buffer pH were classified as WC-CPPs, load flow rate and end of pool collect as KPPs, and operating temperature and bed height as GPPs. No parameter of the step was classified as a CPP.

- F1. With the 3 set-point runs that PCP-005 planned, the confidence interval was wider than the equivalence margin for pool host cell protein, step yield and leached Protein A even at zero difference, so equivalence could not be shown (DEV-005-03). A further 6 set-point runs were made with the margins unchanged. Over all 9 scale-down runs the model is equivalent to manufacturing scale for pool host cell protein and equivalent for peak asymmetry. Leached Protein A is consistently lower in the scale-down model, which is expected from the longer hold of the manufacturing load before capture, giving its host cell proteases more time to cleave ligand. This study does not test that cause. The difference was accepted as a known scale effect, as PCP-005 §5.1 allows. Step yield and column efficiency were not shown equivalent. Their intervals exceed the margin on the low side by 1.4 % and 35 % of the margin (Section 3.1).
- F2. Pool host cell protein rises with protein load and as elution buffer pH falls, and the two parameters interact. It falls as the end of pool collect is extended. The response-surface model for pool host cell protein meets criteria 2 to 4 of PCP-005 §7 and does not meet criterion 5 (finding F7, Section 5.3).
- F3. Step yield rises with end of pool collect and falls with protein load and load flow rate, and flow rate costs yield mainly at high load. The full quadratic model fails the predictive criterion, and a reduced model with the terms significant in both designs meets criteria 2 to 4 of PCP-005 §7. Like every model of the study, it does not meet criterion 5 (Section 5.3).

- F4. No adequate model results for leached Protein A, so its dependence on the parameters of the multivariate design is not resolved. Every run of both designs met its intermediate acceptance criterion of ≤ 158 ppm, with a highest result of 4.61 ppm (Section 5.3).
- F5. Fragments by nrCE-SDS fall slightly as elution buffer pH rises, and no adequate model results. All 28 response-surface pools met the criterion of ≤ 2.45 %, with a highest result of 1.09 % (Section 5.3).
- F6. Under the criteria of Version 2.0 the NOR-propagated PAR for pool host cell protein spans the whole range studied for protein load, load flow rate and end of pool collect. For elution buffer pH it is 3.26–3.9, against a range studied of 3.2–3.9. Every output met its criterion across the NOR (Section 7).
- F7. No residual trend against the predicted value or the run order was found for any response. Between 2 and 8 runs of each response-surface model exceed a Cook's distance of 4/n, so criterion 5 of PCP-005 §7 is not met. Deletion of any one of these runs does not change the conclusion for pool host cell protein (Section 5.5).
- F8. The Protein A eluate held at elution pH met its criteria for 7 days at both 2-8 °C and 21 °C. Aggregate and acidic variants rose more at 21 °C. Fragments by nrCE-SDS rose at both temperatures, and the data do not show a temperature dependence (Section 5.7).
- F9. No difference from the set-point runs was detected at either end of the ranges of operating temperature and bed height, and every result met its criterion. With two runs at each end the intervals are wide (Section 5.6).
- F10. The effect of end of pool collect on pool host cell protein, and its interactions with protein load and elution buffer pH, have opposite signs in the screening and the response-surface designs. The difference is larger than the noise of the screening design, and its cause was not identified (Section 5.3).

Under criterion 1 of PCP-005 §7 a model that fails qualification for an output is not used for that output. The qualification for pool host cell protein supports the PARs and the design space of that output. The scale-down model was not shown equivalent for step yield, so under the same criterion the step yield model is not used for any range or prediction. The KPP classes of load flow rate and end of pool collect rest on the direction of the yield effects, which both designs show, and not on the model (Section 9). The main effects of protein load and elution buffer pH agree in direction with the platform history and the mechanism of the step. Load flow rate did not raise pool host cell protein as the mechanism predicts (Section 5.8).

At commercial scale the integrated model of the train gives a Cpk of 6.14 for host cell protein, 10.76 for fragments and 2,785 for leached Protein A in the drug substance, and no simulated batch of 2,000 fell outside these criteria. The Cpk for leached Protein A is not meaningful, because the simulated drug substance values lie below the limit of quantitation of the ELISA. Three deviations were recorded (DEV-005-01, DEV-005-02 and DEV-005-03). All three were investigated. The affected data were retained, and for DEV-005-03 supplementary runs were made. The results roll up into the Process Characterization Master Report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose primary mechanism of action is antibody-dependent cell-mediated cytotoxicity. It is developed by Novacyte Biologics and is being transferred from development at the sending site to commercial drug substance manufacture at 15,000 L scale at the receiving site, under the Process Transfer Plan PTP-001. The drug substance is purified by the platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small virus retentive filtration and UF/DF.

Protein A chromatography is the first chromatographic step of the purification train. It is operated in bind and elute mode. The resin binds the antibody from the clarified harvest at neutral pH, while host cell protein, DNA and most other components of the feed are removed in the load flow-through and the wash. A low-pH buffer elutes the antibody into a small volume, and the eluate passes to low-pH viral inactivation (Step 6) without neutralization. The step is the principal point of clearance for host cell protein and DNA, and it partly clears fragments. It introduces one attribute of the CQA register, leached Protein A. Although Protein A chromatography removes some virus, no viral clearance claim is made for this step (CMC Biotech Working Group 2009). The modular claim rests on low-pH inactivation, anion exchange and virus filtration, and it is consolidated in PCMR-001.

1.2 Objectives

The objectives of the characterization were the following, as PCP-005 §2 states them.

1. To qualify the scale-down model for the outputs of the step at set-point.
2. To identify which of the 4 parameters of the multivariate design, and which of their two-factor interactions, affect pool host cell protein, step yield and leached Protein A.
3. To fit response-surface models for pool host cell protein, step yield, leached Protein A and fragments by nrCE-SDS, and to judge their adequacy for prediction.
4. To determine the PAR of each multivariate parameter for each attribute with a criterion at the step.
5. To establish the effect of operating temperature and bed height over their ranges, one factor at a time.

6. To establish how long, and at which temperature, the Protein A eluate may be held at elution pH before viral inactivation.
7. To provide the data for the classification of every parameter of the step under SOP-4001, and to support the Stage 2 ranges and the control strategy.

1.3 Regulatory and scientific basis

The study is part of Stage 1 (process design) of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced approach of ICH Q8(R2) and ICH Q11 (International Council for Harmonisation 2009, 2012), in which prior knowledge and risk assessment identify the parameters that may affect quality, and designed experiments then measure and bound those effects. Criticality is treated as a continuum, as ICH Q8(R2) describes it, and risk is managed under ICH Q9 (International Council for Harmonisation 2023). The classification of parameters follows the decision logic of the A-Mab case study (CMC Biotech Working Group 2009), as PCMP-001 §7.7 applies it. The documents this report depends on and feeds are listed below.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-005	Process Characterization Plan (Protein A Chromatography (Step 5))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision

1.4 Changes in this revision

Version 1.0 of this report was approved on 2026-07-24. Version 2.0 takes effect on 2026-09-30 and is issued under INV-001. QA opened the investigation on 2026-08-03 after its review of PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18 with the classification major. The investigation found that the criteria used to judge each step in the PAR analysis were not in any approved protocol and had been set after the data were seen (INV-001 finding 1). For this step, Version 1.0 judged pool host cell protein against ≤ 10,724 ng/mg. That limit was the drug substance

Table 1.2: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0.

Step	Attribute	Version 1.0[b]		Version 2.0	
		Criterion	Rule	Criterion	Basis
Protein A	Leached Protein A (ppm)	none[a]		≤ 158	back-calculated
	Fragments, LMW (%)	not assessed[c]		≤ 2.45	back-calculated
	Host cell protein (ng/mg)	≤ 10,724	[R1] margin 4.5	≤ 16,000	back-calculated

[a] Version 1.0 set no step criterion and judged the step against the drug-substance criterion.
[b] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[c] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
[R1] Version 1.0 rule: nominal back-calculation / margin.

criterion carried back through the nominal downstream clearance of this campaign and divided by a margin of 4.5 (INV-001 finding 2). Leached Protein A was judged against the drug substance criterion, and fragments were not reported. Table 1.2 sets the Version 1.0 limits beside the Version 2.0 criteria.

The Version 2.0 criteria are intermediate acceptance criteria derived a priori by PCMP-001 §7.3. The criterion for pool host cell protein, ≤ 16,000 ng/mg, is wider than the Version 1.0 limit. The Version 1.0 limit was never approved and appears in no approved plan, and it is replaced by a criterion derived without data from the studies it judges. The rule that data may tighten a criterion and never loosen it applies to the Version 2.0 criteria from this revision on.

The experiments, the data and the fitted models are those of Version 1.0. What moved is the following.

1. The NOR-propagated PAR for pool host cell protein. Under the Version 1.0 limit it was 10–34.5 g/L resin for protein load, which ends inside the NOR of 20–40 g/L resin and made the upper end of the NOR appear unsupported. Under Version 2.0 it is 10–50 g/L resin. For elution buffer pH it moved from 3.48–3.9 to 3.26–3.9, for load flow rate from 138–300 to 100–300 cm/hr, and for end of pool collect from 2.29–3.2 to 2–3.2 CV.
2. The NOR-propagated PAR of elution buffer pH for leached Protein A. Under the drug substance criterion of ≤ 5 ppm it was 3.2–3.86. Leached Protein A is now judged against its intermediate acceptance criterion and reported descriptively, because no adequate model results for it (Section 5.3).
3. Fragments by nrCE-SDS are reported for the first time. The retained response-surface pools and the hold samples of days 0 and 7 were tested in study S-PA-RET under AMD-001 (Table 3.3). The SEC fragment results of both designs are kept in the record and are not used for any decision (INV-001 finding 6).
4. Each response-surface model is judged against the model adequacy criteria of PCP-005 §7, which Version 1.0 left to the analysis. The full quadratic model for step yield was replaced by a reduced model, and leached Protein A and fragments are reported descriptively (Section 5.3).

5. The scale-down model qualification is reported against the equivalence test of PCP-005 §5.1. The planned set-point runs could not demonstrate equivalence whatever the means were, and 6 supplementary set-point runs (study S-PA-SDM2) were made under DEV-005-03, with the margins unchanged (Section 3.1).
6. The execution window of every study is recorded (Table 4.3, INV-001 finding 5).

The parameter effects and their directions are those of Version 1.0. Their discussion is extended in this revision (Section 5.3). Acidic variants are not an output of this step, so the two-sided capability assessment of INV-001 finding 4 does not apply here.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The same capture step has been used to purify the platform antibodies X-Mab, Y-Mab and Z-Mab, and A-Mab through clinical manufacture. Their history establishes the mechanism of the step and the direction of its main effects.

Staphylococcal Protein A, immobilized on the resin, binds the Fc region of the antibody at the interface between the second and third constant domains of the heavy chain. When the pH is lowered, histidine residues at this interface become protonated, the interaction weakens, and the antibody is released. The expectations for each output follow from this mechanism.

Host cell protein reaches the eluate by two routes. Some binds weakly to the resin backbone or the ligand and is released when the pH drops, and some is associated with the antibody and is carried through as a complex. The host cell protein associated with the bound antibody is carried on the bed with it, so its amount rises with the mass of antibody loaded. A lower elution pH releases more of the weakly bound host cell protein with the product. Pool host cell protein was therefore expected to rise with protein load and as elution buffer pH falls. At high load more host cell protein is held weakly on the bed, so a low elution pH has more to release. An interaction between the two parameters was expected for this reason.

Step yield is lost when antibody breaks through during the load or remains on the column at the end of collection. Binding is limited by diffusion into the pores of the bead. For this reason breakthrough rises as the load approaches the dynamic binding capacity and as the residence time falls. A higher load flow rate shortens the residence time and lowers the dynamic binding capacity, and this costs yield only when the bed is close to capacity. Yield was therefore expected to fall with flow rate mainly at high load, and to rise as the end of pool collect is extended into the tail of the elution peak.

Leached Protein A comes from hydrolysis of the coupling chemistry and from cleavage of exposed ligand by host cell proteases in the load, and the leached fragments elute with the antibody because they still bind Fc. Leaching is highest on new resin. More protease activity in the load, for example after a longer hold of the clarified harvest before capture, is expected to cleave more ligand. Neither of these is a setting of the run, so within the ranges of one run leached Protein A was expected to be flat.

A fragment is captured only if it still carries Fc. Fab fragments flow through with the load, and the pool carries less fragment than the feed. A lower elution pH may add a little acid-catalysed hinge hydrolysis during elution. The pool also still carries host

cell protein, part of which elutes in the low molecular weight window of SEC and is read as fragment. SEC therefore overstates fragments in this matrix, and nrCE-SDS does not (INV-001 finding 6). DNA is polyanionic and does not bind Protein A. Its clearance is large and was not expected to depend on the operating parameters within their ranges.

Protein A, when operated under platform conditions, does not separate glycan variants or charge variants, because it binds the Fc region independently of both (CMC Biotech Working Group 2009). Nor does it separate aggregate from monomer, since both carry Fc. In the Protein A characterization of the case study no process parameter had a significant effect on aggregate (CMC Biotech Working Group 2009). Aggregate formation at low pH over the inactivation hold is characterized in PCR-006. For these reasons glycan variants, charge variants and aggregate were not responses of the designs.

2.2 Quality attributes in scope

Leached Protein A is the only attribute of the CQA register that this step introduces. The step is also the principal point of clearance for host cell protein and DNA, and it partly clears fragments. Table 2.1 lists the attributes in scope with their drug substance criteria and the criticality level the register assigns. The register treats criticality as a continuum, as the A-Mab case study does (CMC Biotech Working Group 2009). Each attribute is a CQA with a level from very low (VL) to very high (VH), assigned with Tool #1, whose score is the product of impact and uncertainty.

Table 2.1: Quality attributes in scope of this report, with the drug substance criterion and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Leached Protein A	process impurity	≤ 5 ppm	L-M	16
Host Cell Protein (HCP)	process impurity	≤ 100 ng/mg	M-H	36
Fragments (LMW, clipped species)	size variant	≤ 2 % (nrCE-SDS)	L	12
Residual DNA	process impurity	≤ 0.001 ng/dose	VL	6
Aggregates (HMW)	size variant	≤ 5 % HMW (SEC)	H	60

The drug substance criteria do not apply to the Protein A pool directly. Cation exchange and anion exchange remove host cell protein, leached Protein A and fragments after this step, and judging the pool against the drug substance limit would fail every acceptable run. Each output of the step with a downstream dependence

is therefore judged against an intermediate acceptance criterion (Section 7). Residual DNA and aggregate were measured for information. Step yield is a performance attribute without an acceptance criterion, and it is used for the classification of parameters that act on process performance.

2.3 Risk-based prioritization and parameter selection

RA-001 assessed every parameter of the step before characterization began. Each parameter was ranked on its potential impact on the CQAs and on process performance, and on its potential to interact with other parameters. RA-001 gives the scores and the rationale.

Protein load, elution buffer pH, load flow rate and end of pool collect were assigned to multivariate study. Each carries an effect expected from the mechanism, and the interactions of protein load with elution buffer pH and with load flow rate can only be estimated in a design that varies the parameters together. Operating temperature and bed height were assigned to univariate study. A higher temperature speeds the diffusion of the antibody into the pores, which raises the dynamic binding capacity, and it strengthens the hydrophobic contacts at the Fc–ligand interface, which makes elution slightly less complete. Over the range studied both effects were expected to be small against those of load and pH, and the resin manufacturer specifies binding capacity across that range. Bed height is held by the packing specification in manufacture, and at constant linear velocity it sets the residence time.

The ranges studied are between 1.9 and 2.3 times the width of the NOR. They show how the step behaves near and beyond the edges of routine operation, and they bound the knowledge space within which a PAR can be claimed.

3 Materials and methods

3.1 Scale-down model and its qualification

The scale-down model is a laboratory column packed with the commercial Protein A resin and operated under SOP-2007 on the scaling principles of PCMP-001 §5. The bed height and the residence time of the commercial column were kept. Load, wash and elution volumes were normalized to column volumes, and buffers were made to the commercial recipes. Column efficiency was checked by plate count and peak asymmetry after packing, and both are compared with manufacturing scale below.

The model was qualified under SOP-1001 in study S-PA-SDM (2025-11-05 to 2025-11-14), before the first run of the screening design, and in the supplementary study S-PA-SDM2 (2026-09-07 to 2026-09-11) made under DEV-005-03. Set-point runs of the scale-down model were compared with the 5 manufacturing-scale batches of A-Mab run at set-point. The outputs compared were pool host cell protein, step yield, leached Protein A, and column efficiency as plates per metre and peak asymmetry. Equivalence of the means was tested by two one-sided tests at $\alpha = 0.05$, using a Student t interval with pooled variance. The equivalence margin for each output is half the range of the manufacturing-scale batches. The model is qualified for an output when the 90 % confidence interval of the difference in means lies within that margin.

With the 3 runs of S-PA-SDM the half-width of the interval alone was larger than the margin for pool host cell protein, step yield and leached Protein A, so equivalence could not be shown for them whatever the means were (DEV-005-03, Section 13). A further 6 set-point runs were therefore made. The margins were not changed. Table 3.1 gives the result over all 9 scale-down runs, with 12 degrees of freedom.

Table 3.1: Scale-down model qualification over all scale-down set-point runs (studies S-PA-SDM and S-PA-SDM2). SD, scale-down model. M, manufacturing scale. Difference and 90 % confidence interval (CI), scale-down minus manufacturing. Step yield in percentage points.

Output	n (SD / M)	Mean, SD	Mean, M	Difference	90 % CI	Margin	Equivalent
Pool HCP (ng/mg)	9 / 5	7,712	7,964	-251	-929 to 427	± 1,026	Yes
Step yield (%)	9 / 5	94.18	94.42	-0.24	-2.14 to 1.67	± 2.11	No
Leached Protein A (ppm)	9 / 5	2.52	3.70	-1.18	-1.97 to -0.39	± 1.08	No

Output	n (SD / M)	Mean, SD	Mean, M	Difference	90 % CI	Margin	Equivalent
Column efficiency (plates/m)	9 / 5	2,533	2,649	-116	-320 to 89	± 237	No
Peak asymmetry	9 / 5	1.20	1.22	-0.02	-0.09 to 0.05	± 0.12	Yes

The scale-down model met the equivalence criterion for 2 of the 5 outputs, pool host cell protein and peak asymmetry.

For pool host cell protein the difference in means was -251 ng/mg, and the interval of -929 to 427 ng/mg lies within the margin of ± 1,026 ng/mg. The model is qualified for pool host cell protein.

For leached Protein A the scale-down mean was 2.52 ppm against 3.70 ppm at manufacturing scale, and the whole confidence interval lies below zero. Each scale-down run was below the manufacturing mean. PCP-005 §5.1 accepts a difference that is consistent in direction and explained by a known scale effect, provided this report carries it into the interpretation. The difference is consistent in direction, and a scale effect on leached Protein A is recorded for this step. Its expected cause is the longer hold of the clarified harvest before capture at manufacturing scale, which gives host cell proteases more time to cleave ligand from the resin. This study does not test that cause. The difference was accepted as a known scale effect. It is carried into the interpretation of leached Protein A in Section 5.3, where the design results are scaled by the ratio of the manufacturing mean to the scale-down mean, 1.47.

For step yield the difference in means was -0.24 percentage points. The interval of -2.14 to 1.67 points extends beyond the margin of ± 2.11 on the low side, by 1.4 % of the margin, so equivalence was not shown. No known scale effect applies to step yield. Under criterion 1 of PCP-005 §7 the step yield model is therefore not used for any range or prediction. The yield effects reported in Section 5.3 describe the scale-down model. Their direction, which both designs show, supports the KPP classes in Section 9, and their size at manufacturing scale is not established.

For column efficiency the scale-down column averaged 2,533 plates per metre against 2,649 at manufacturing scale. The interval of -320 to 89 extends beyond the margin of ± 237 on the low side, by 35 % of the margin, so equivalence was not shown. Peak asymmetry was equivalent. Column efficiency is a check of the packing and not a response of the characterization, so no result of this report depends on it directly. A lower plate count would broaden the peaks of the laboratory column. Pool host cell protein, which depends on how sharply the product elutes, was nevertheless shown equivalent.

Fragments by nrCE-SDS were not part of the qualification, because the analytical method decision AMD-001 was taken after the model was qualified. The model is taken as representative for fragments on mechanism. Capture of Fc-bearing species and low-pH hinge hydrolysis do not depend on scale at equal bed height, residence time and elution pH.

3.2 Operation

Each run followed SOP-2007 at laboratory scale. The column was equilibrated, loaded with clarified harvest to the target protein load, washed, and eluted with the elution buffer at the pH set by the design. The eluate was collected from the rise of the absorbance at 280 nm until the end of pool collect set by the design, in column volumes. The column was then stripped, cleaned and stored under SOP-2008. The pool was not neutralized, because the commercial eluate passes to low-pH viral inactivation at elution pH.

Each design was loaded from one pool of clarified harvest produced at set-point and held frozen in aliquots. A single feed keeps feed variation out of the estimated effects. The effect of feed quality on the step was therefore not estimated in these studies. The case study reports that Protein A capture is robust over a wide range of feed inputs (CMC Biotech Working Group 2009), and the feed is characterized in PCR-003 and PCR-004. Runs were executed in random order within each design. The resin was used within the life-cycle limits of SOP-2008, and the life cycle of the resin is outside the scope of this report.

3.3 Analytical methods

Each response was measured by a validated method under its own AMV. The controlled procedures and method validations used are listed below. The SOP and AMV numbers are placeholders.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
SOP-3015	Purity by Non-Reduced CE-SDS	SOP
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

Table 3.3: Methods used for size variants, by step and study.

Step	Attribute	Method used in study	
		Screening	Response surface
Protein A	Aggregates, HMW	SEC	SEC
	Fragments, LMW	SEC[a)]	SEC[a)], nrCE-SDS[b)]

[a)] In this matrix part of the host cell protein elutes in the SEC low molecular weight window. Reported for information only (AMD-001).
[b)] Retained response-surface samples, tested 2026-04-06 to 2026-04-17.

Reference	Title	Type
AMV-3019	Protein Concentration by A280 (UV)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3020	Purity (non-reduced CE-SDS)	Method validation

Pool host cell protein was measured by ELISA (AMV-3012) and is reported per mass of antibody in the pool, so that it does not depend on how far the pool is diluted by the end of pool collect. Step yield is the mass of antibody in the pool divided by the mass loaded, from the protein concentration by absorbance at 280 nm (AMV-3019) and the measured volumes. Leached Protein A was measured by ELISA (AMV-3016). Fragments were measured by nrCE-SDS (AMV-3020, under SOP-3015), as PCMP-001 §7.8 sets for every step. SEC (AMV-3011) remains the method for aggregate, where no co-elution of this kind applies. Acidic variants in the hold study were measured by icIEF (AMV-3013), and residual DNA by qPCR (AMV-3014). Table 3.3 shows which method measured fragments in which study of the step, and Appendix D summarizes the methods.

The screening design ran before AMD-001, and its pools were not retained. Its fragment results are by SEC only and are not used for any decision. The response-surface pools were retained frozen and were tested by nrCE-SDS in study S-PA-RET (2026-04-06 to 2026-04-17). On those pools SEC read fragments higher than nrCE-SDS by 0.12 to 1.37 percentage points (median 0.36), which is the bias that INV-001 finding 6 describes for a matrix that still carries host cell protein.

3.4 Sampling plan

The load of each design was sampled once per feed aliquot thawed. The pool of every run was sampled once at the end of collection, after mixing, and its volume was recorded. Each pool sample was divided into aliquots for each method. One further aliquot of every response-surface pool was retained frozen under the sample

management procedure of Analytical Development, and these are the samples tested by nrCE-SDS in study S-PA-RET. The screening design had 3 centre points and the response-surface design 4. Residual DNA was measured on the load and pool of the centre-point runs. The hold study was sampled as described in Section 4.5.

3.5 Statistical methods

The designs were built and analysed under SOP-1002 with the common approach of PCMP-001 §6. Factors were coded from −1 to +1 over their range studied, with 0 at the centre. For each of the 4 design factors the centre of the range studied is the set-point, so the centre-point runs of both designs are set-point runs.

The screening data were fitted with a model of main effects and two-factor interactions. Effects are reported as the change in the response over the coded range of the factor, which is twice the coefficient. Significance was judged at $\alpha = 0.05$. The screening model identifies effects. A two-level design cannot resolve curvature, so the screening model was not used to predict between its corners.

The response-surface data were fitted with a full quadratic model. This is the predictive model of the step, and every PAR and every prediction of a quality attribute in this report comes from it. Step yield is the exception, because its full quadratic model failed criterion 4 and was replaced by a reduced model (Section 5.3). Coefficients are reported in coded units, so that each measures the change in the response over half the range studied. Effects quoted from them are twice the coefficient. Each model is reported with R^2, adjusted R^2 and predicted R^2 (from PRESS), the model F-test, and the test of lack of fit against pure error from the centre points. The adequacy criteria of PCP-005 §7 were applied as follows.

The numbers are those of PCP-005 §7, where criterion 1 is the qualification of the scale-down model (Section 3.1).

2. Model significance. The model F-test is significant at $\alpha = 0.05$.
3. Lack of fit. The lack of fit test is not significant at $\alpha = 0.05$.
4. Predictive power. Predicted R^2 falls short of R^2 by no more than 0.20.
5. Residuals. Regressions of the absolute residuals on the predicted values, and of the residuals on run order, show no slope significant at $\alpha = 0.05$, and no run has a Cook’s distance above 4/n (0.14 for 28 runs).

A model that fails criterion 3 or 4 was refitted with a justified change of terms, or it was not used to set a range. Where no adequate model resulted, the response is reported descriptively against its criterion over the runs of the design.

The PAR analysis follows PCMP-001 §6.4 and is described in Section 7. Capability at commercial scale was estimated from the integrated model of the train, which runs 2,000 simulated batches with each parameter drawn within its NOR. Capability is reported as a one-sided Cpk against the drug substance criterion for each attribute of

this step, and is judged against the target of 1.33 and the minimum of 1.00 that PCMP-001 §7.9 sets, and the same simulation gives the probability of an out-of-specification drug substance.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 lists the parameters of the step with their set-point, NOR, range studied, study type and the classification proposed in Section 9.

Table 4.1: Parameters of the Protein A step with set-point, NOR, range studied (Char. range), classification and study type.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Protein load	g/L resin	30	20–40	10–50	WC-CPP	multivariate
Elution buffer pH	pH	3.55	3.4–3.7	3.2–3.9	WC-CPP	multivariate
Load flow rate	cm/hr	200	150–250	100–300	KPP	multivariate
End of pool collect	CV	2.6	2.3–2.9	2–3.2	KPP	multivariate
Operating temperature	°C	22	18–26	15–30	GPP	univariate
Bed height	cm	20	15–25	10–30	GPP	univariate

Load flow rate is set as a linear velocity, so at the set-point bed height it fixes the residence time. The residence time is 6 min at the set-point flow rate, and over the range studied it runs from 4 min at the highest flow to 12 min at the lowest. The upper end of the flow range therefore tests the shortest residence time against the highest load, which is the condition most likely to cause breakthrough. The lower end of the elution pH range tests the strongest elution and the largest release of weakly bound host cell protein. The upper end tests the weakest elution, with the broadest peak and the least co-eluting host cell protein. Table 4.2 gives the letter code of each design factor, which the coefficient tables and the appendices use.

Table 4.2: Letter codes of the design factors.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Elution buffer pH	pH	screening + RSM

Code	Factor	Unit	Studied in
C	Load flow rate	cm/hr	screening + RSM
D	End of pool collect	CV	screening + RSM

The execution window of every study of the step is recorded in Table 4.3. Each study began after PCP-005 Version 1.0 was approved on 2025-11-03.

Table 4.3: Execution window of each study of the Protein A step.

Study	Study type	Start	End
S-PA-SDM	scale-down model qualification	2025-11-05	2025-11-14
S-PA-UNI	univariate study	2025-12-08	2025-12-12
S-PA-SDM2	scale-down model qualification	2026-09-07	2026-09-11
S-PA-SCR	screening design	2025-11-17	2025-12-05
S-PA-RSM	response-surface design	2026-01-12	2026-02-06
S-PA-HLD	hold study	2026-02-02	2026-02-16
S-PA-RET	retest of retained samples	2026-04-06	2026-04-17

4.2 Screening design

The screening design (study S-PA-SCR, 2025-11-17 to 2025-12-05) was a full two-level factorial design in the 4 factors, with 16 factorial runs and 3 centre points, 19 runs in all. Because the design is a full factorial, no effect is aliased with another, and every main effect and every two-factor interaction can be estimated. The responses were pool host cell protein, step yield and leached Protein A, with fragments by SEC recorded for information. All 4 factors were carried into the response-surface design whatever the screening result, as PCP-005 §6.2 planned. The screening design therefore served to identify effects and interactions and to check the ranges. Appendix A gives the matrix and the results.

4.3 Response-surface design

The response-surface design (study S-PA-RSM, 2026-01-12 to 2026-02-06) was a face-centred central composite design in the same 4 factors. It has 16 factorial runs, 8 axial runs at the faces of the cube and 4 centre points, 28 runs in all. A face-centred design keeps every run inside the range studied. The full quadratic model leaves 13 residual degrees of freedom, with 3 for pure error and 10 for lack of fit. The responses were pool host cell protein, step yield, leached Protein A and fragments by nrCE-SDS on the retained pools. Appendix B gives the matrix and the results.

4.4 Univariate assessment

Operating temperature and bed height were studied one factor at a time in study S-PA-UNI (2025-12-08 to 2025-12-12), with the other parameters at their set-points. Each was run 2 times at the low and the high end of its range studied, and the 3 set-point runs of study S-PA-SDM served as the reference, as PCP-005 §6.4 planned. Bed height was varied at the set-point linear velocity, so that the residence time changed with it. The shortest bed gives a residence time of 3 min, shorter than any run of the designs. The responses were pool host cell protein, step yield and leached Protein A. Fragments were not measured in this study, which ran before AMD-001. A univariate study cannot detect an interaction. The ranges of these two parameters are therefore supported only with the other parameters at their set-points.

4.5 Hold study of the Protein A eluate

The Protein A eluate is held at elution pH before viral inactivation. A pool produced at set-point was divided and held at 2-8 °C and 21 °C for up to 7 days in study S-PA-HLD (2026-02-02 to 2026-02-16). Samples were taken on days 0, 1, 3, 5 and 7 at each temperature. Aggregate and fragments were measured by SEC, and acidic variants by icIEF, at every sample point. The samples of days 0 and 7 were retained and tested by nrCE-SDS in study S-PA-RET. Because Protein A capture does not separate aggregate or charge variants, the held eluate carries the criteria that PCMP-001 sets for the output of the production bioreactor, ≤ 6.83 % HMW and 18–39 % acidic variants. Fragments are judged against the intermediate acceptance criterion of the pool, ≤ 2.45 % by nrCE-SDS, on the days with an nrCE-SDS result.

5 Results

5.1 Centre-point performance and reproducibility

The centre points of each design were run at set-point and estimate the pure error against which lack of fit is tested. Table 5.1 gives their mean, standard deviation and coefficient of variation in the response-surface design, and Table 5.2 in the screening design.

Table 5.1: Centre-point reproducibility in the response-surface design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	7.56e+03	357	4.72
Step yield	4	0.946	0.0187	1.97
Leached Protein A (ppm)	4	2.86	0.967	33.8

Table 5.2: Centre-point reproducibility in the screening design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	3	8.1e+03	1.61e+03	19.9
Step yield	3	0.935	0.00599	0.64
Leached Protein A (ppm)	3	3.28	0.508	15.5

Step yield was the most reproducible response, with a coefficient of variation of 1.97 % in the response-surface design. Pool host cell protein reproduced to 4.72 % in the response-surface design and to 19.88 % in the screening design. The larger screening value makes the screening tests for pool host cell protein less sensitive than those of the response-surface design. Leached Protein A was the least precise response, at 33.84 % in the response-surface design. Its centre-point spread is of the same size as its variation across the whole design, so the variation of the ELISA and of the runs at set-point masks any effect of the parameters of that size.

5.2 Screening: factor effects

Table 5.3 summarizes the screening models. The models for pool host cell protein and step yield are significant. The model for leached Protein A is not ($p = 0.867$), and none of its terms is significant.

Table 5.3: Screening model summary (main effects and two-factor interactions).

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.973	0.939	0.886	28.6	3.49e-05	1.4e+03
Step yield	19	0.976	0.945	0.798	32	2.28e-05	0.00833
Leached Protein A (ppm)	19	0.371	-0.414	-2.98	0.473	0.867	0.513

Pool host cell protein was affected by elution buffer pH (B) and protein load (A), and by their interaction (AB). Raising the elution pH across its range lowered pool host cell protein by 8,840 ng/mg, and raising protein load across its range raised it by 7,269 ng/mg. The AB interaction (-2,587 ng/mg, p = 0.006) is negative, so the fall in host cell protein with rising pH is steeper at high load. Load flow rate and end of pool collect had no significant effect in the screening design. The effect of end of pool collect was positive in this design (708 ng/mg, p = 0.341), the opposite sign to the response-surface design (Section 5.3). Table 5.4 gives the effects.

Table 5.4: Screening effects on pool HCP (ng/mg), sorted by size.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8,840.3	-4,420.2	350	-12.6	1.45e-06	***
A	7,269.4	3,634.7	350	10.4	6.4e-06	***
AB	-2,586.7	-1,293.4	350	-3.7	0.00608	**
AD	1,132.9	566.4	350	1.62	0.144	
AC	830.1	415.1	350	1.19	0.27	
D	707.9	353.9	350	1.01	0.341	
BD	-433.1	-216.5	350	-0.619	0.553	
C	167.3	83.7	350	0.239	0.817	
CD	-70.6	-35.3	350	-0.101	0.922	
BC	6.1	3.0	350	0.00868	0.993	

Step yield was affected by end of pool collect (D), protein load (A) and load flow rate (C), and by the AC interaction. Extending the end of pool collect across its range raised yield by 6.0 percentage points. Raising protein load and load flow rate lowered it by 2.6 and 2.4 percentage points. The AC interaction (-2.1 percentage points, p = 0.001) means that flow rate costs more yield at high load. The CD and BD interactions were also significant, at -1.1 and -1.0 percentage points. These are smaller than the main effects, and they were not confirmed in the response-surface design. Elution buffer pH had no significant effect on yield. Table 5.5 gives the effects, and Appendix C gives the effects for leached Protein A.

Table 5.5: Screening effects on step yield (fraction), sorted by size.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.0604	0.0302	0.00208	14.5	5.01e-07	***
A	-0.0255	-0.0128	0.00208	-6.13	0.00028	***
C	-0.0237	-0.0118	0.00208	-5.68	0.000463	***
AC	-0.0209	-0.0104	0.00208	-5.01	0.00104	**
CD	-0.0107	-0.00537	0.00208	-2.58	0.0326	*
BD	-0.00981	-0.0049	0.00208	-2.36	0.0463	*
AB	0.00443	0.00221	0.00208	1.06	0.319	
AD	0.0037	0.00185	0.00208	0.889	0.4	
BC	0.0035	0.00175	0.00208	0.84	0.425	
B	-0.00236	-0.00118	0.00208	-0.568	0.586	

The screening results agree with the expectations of the platform history in direction for protein load and elution buffer pH and for the AB interaction on pool host cell protein, and for every main effect and the AC interaction on step yield. This type of design cannot attribute curvature to a factor, and the effects were refined in the response-surface design.

5.3 Response-surface models

Table 5.6 judges each response-surface model against criteria 2 to 4 of PCP-005 §7. Criterion 5, on the residuals, is reported separately in Section 5.5.

Table 5.6: Adequacy of the response-surface models against PCP-005 §7, criteria 2 to 4.

Response	Model	R^2	Adj. R^2	Pred. R^2	R^2 − pred. R^2	Model p	Lack of fit p	Criteria 2–4 met
Pool HCP (ng/mg)	full quadratic	0.989	0.976	0.932	0.057	< 0.001	0.058	Yes
Step yield	full quadratic	0.913	0.818	0.586	0.326	< 0.001	0.731	No
Step yield	reduced (5 terms)	0.831	0.801	0.759	0.072	< 0.001	0.693	Yes
Leached Protein A (ppm)	full quadratic	0.567	0.101	-0.714	1.28	0.365	0.953	No
Fragments (LMW, %, nrCE-SDS)	full quadratic	0.599	0.168	-1.32	1.92	0.280	0.644	No

The model for pool host cell protein meets criteria 2 to 4. The full quadratic model for step yield is significant and shows no lack of fit, but its predicted R^2 falls short of R^2 by 0.326, more than the limit of 0.20. It was therefore refitted with the terms that were significant in both designs, the main effects of A, C and D and the AC interaction. The reduced model meets criteria 2 to 4. The models for leached Protein A and fragments by nrCE-SDS are not significant ($p = 0.365$ and 0.280), and both have a negative predicted R^2. No adequate model results for them, and they are reported descriptively. The same verdicts follow when predicted R^2 is compared with adjusted R^2, as PCMP-001 §7.2 states the criterion.

5.3.1 Pool host cell protein

Table 5.7 gives the coefficients of the model for pool host cell protein, and Table 5.8 its analysis of variance.

Table 5.7: Response-surface coefficients for pool HCP (ng/mg), coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	7,215.5	309	23.3	5.44e-12	***
A	3,871.8	211	18.4	1.12e-10	***
B	-5,244.6	211	-24.9	2.4e-12	***
C	-435.4	211	-2.06	0.0596	
D	-1,055.5	211	-5	0.000242	***
AB	-1,899.8	224	-8.49	1.16e-06	***
AC	-429.1	224	-1.92	0.0774	
AD	-518.1	224	-2.32	0.0376	*
BC	348.2	224	1.56	0.144	
BD	727.4	224	3.25	0.00632	**
CD	-128.7	224	-0.575	0.575	
A^2	-344.5	557	-0.618	0.547	
B^2	1,943.5	557	3.49	0.00401	**
C^2	851.3	557	1.53	0.151	
D^2	330.5	557	0.593	0.563	

Table 5.8: Analysis of variance with lack of fit for pool HCP (ng/mg).

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.08e+08	14	6.48e+07	80.9	2.71e-10
Residual	1.04e+07	13	8.01e+05		
Lack of fit	1e+07	10	1e+06	7.88	0.058
Pure error	3.82e+05	3	1.27e+05		

The response-surface model confirms the screening result for protein load and elution buffer pH and refines it. Elution buffer pH (-5,245 ng/mg per half range) and protein

load (3,872 ng/mg) carry the largest coefficients, and the AB interaction is significant (-1,900 ng/mg, $p < 0.001$). At high load the pH coefficient is -7,144 ng/mg per half range, and at low load it is -3,345 ng/mg. The B^2 term is significant and positive (1,944 ng/mg), so pool host cell protein rises more steeply at the low end of the pH range. The lack of fit test ($p = 0.058$) is close to its limit.

End of pool collect lowers pool host cell protein. Its coefficient is -1,055 ng/mg per half range ($p < 0.001$), an effect of -2,111 ng/mg across the range studied. The AD interaction (-518 ng/mg) makes the fall steeper at high load, and the BD interaction (727 ng/mg) makes it shallower at high pH. Collecting further into the tail of the elution peak adds antibody with little host cell protein, because the weakly held impurity elutes early in the low-pH transition. The host cell protein per mass of antibody therefore falls, and it falls most where there is most impurity in the early part of the peak, at high load and low pH.

These three terms reverse sign between the designs. In the screening design the effect of end of pool collect was 708 ng/mg, and the AD and BD effects were 1,133 and -433 ng/mg, against -2,111, -1,036 and 1,455 ng/mg in the response-surface design. None of the screening estimates is significant, and each has a standard error of 700 ng/mg. Noise alone does not explain the reversal, however. When the screening results are compared with the predictions of the response-surface model, the difference depends on end of pool collect by 2,819 ng/mg across its range ($p = 0.020$). Curvature does not explain it either, because in a full factorial design the quadratic terms are confounded with the intercept and not with the main effect of end of pool collect. The cause was not identified (finding F10). The two designs were run at different times (Table 4.3), and a difference between their feeds or in how the end of collection was set could act in this way. The response-surface model is the predictive model, and the effect of end of pool collect is taken from it.

Load flow rate has no significant effect on pool host cell protein ($p = 0.060$), and the AC interaction is not significant either ($p = 0.077$). Both point estimates are negative. The mechanism predicted a small rise in pool host cell protein with a faster load at high load, and the data do not support that prediction.

Across the response-surface design pool host cell protein ranged from 2,149 to 25,114 ng/mg. In 8 runs of the two designs it exceeded the intermediate acceptance criterion of ≤ 16,000 ng/mg. Each of these runs was at the highest protein load and the lowest elution pH of the range studied, outside the NOR. Figure 5.1 shows the response surfaces over protein load and elution buffer pH, with load flow rate and end of pool collect at their set-points.

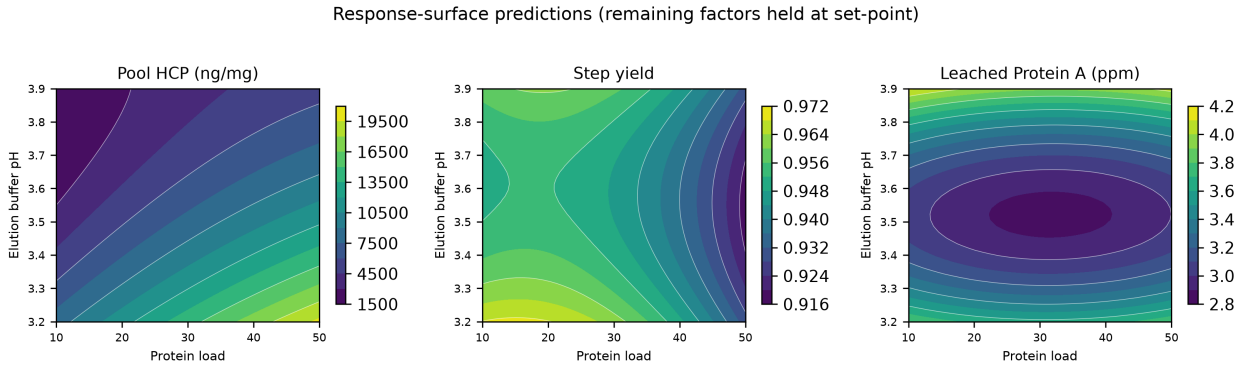


Figure 5.1: Response surfaces over protein load and elution buffer pH, with load flow rate and end of pool collect at their set-points (the design centre). Leached Protein A has no adequate model and is shown for information.

Figure 5.2 shows the diagnostics of the model. The residuals scatter evenly about zero across the predicted range, and they fall close to the line on the normal probability plot.

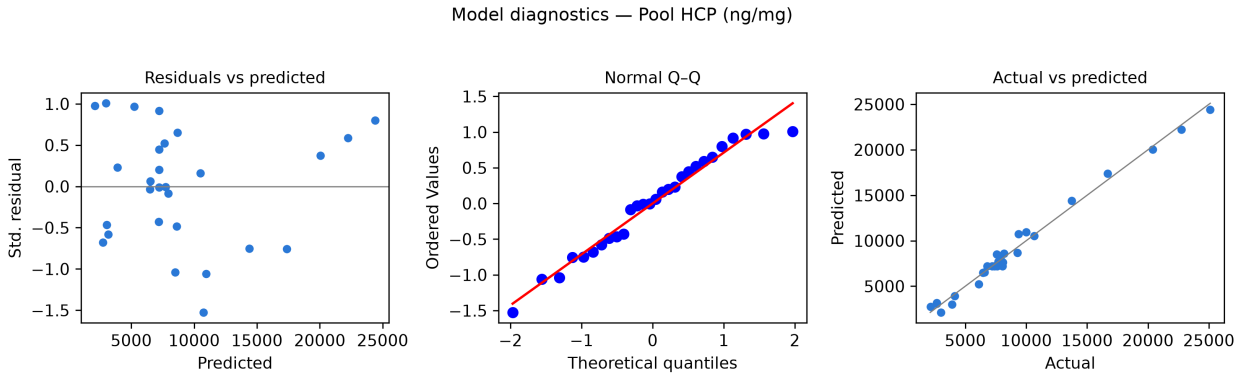


Figure 5.2: Diagnostics of the response-surface model for pool HCP (ng/mg).

5.3.2 Step yield

The reduced model for step yield has an intercept of 94.0 %. Its coefficients per half range are 3.7 percentage points for end of pool collect, -1.6 for protein load, -0.9 for load flow rate and -0.9 for the AC interaction. At high load the coefficient of flow rate is -1.8 percentage points per half range. At low load it is 0.0, so flow rate has almost no effect on yield when the bed has capacity to spare. Across the response-surface design step yield ranged from 84.4 to 99.0 %. Step yield has no acceptance criterion, and its model is used for the classification of parameters only. The coefficients of the full quadratic model are given in Appendix C.

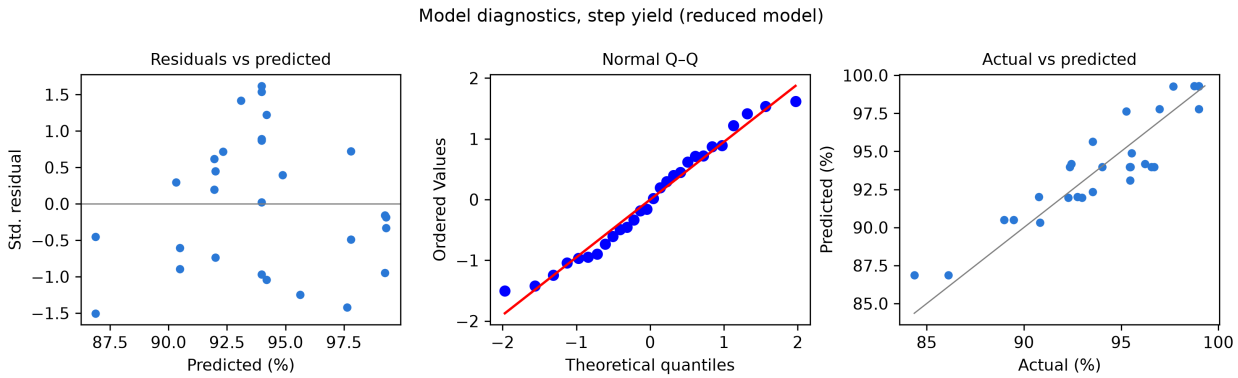


Figure 5.3: Diagnostics of the reduced response-surface model for step yield (A, C, D and AC).

5.3.3 Leached Protein A

The leached Protein A model is not significant as a whole. One of its 14 terms, B^2, is significant, which is about what chance alone gives at $\alpha = 0.05$, and the centre points vary by 34 %. A model this noisy cannot show that leached Protein A is independent of the parameters. It shows only that any dependence is too small to resolve against the variation of the runs. Leached Protein A ranged from 1.70 to 4.61 ppm across the runs of both designs, against an intermediate acceptance criterion of ≤ 158 ppm. Scaled by the factor of 1.47 between manufacturing and scale-down means (Section 3.1), the highest result corresponds to 6.77 ppm at manufacturing scale, which still meets the criterion (finding F4).

5.3.4 Fragments by nrCE-SDS

Fragments by nrCE-SDS ranged from 0.77 to 1.09 % across the 28 retained pools of the response-surface design, against an intermediate acceptance criterion of ≤ 2.45 %. The model as a whole is not significant. Within it, elution buffer pH has a significant negative coefficient (-0.053 % per half range, $p = 0.008$), so fragments fall slightly as elution pH rises. The size of the effect is small against the criterion. Every pool met the criterion. No PAR is claimed from the model (finding F5).

5.4 Attributes measured for information

Aggregate by SEC was measured on every pool of both designs, for information. It ranged from 1.40 to 1.73 % HMW across the 47 pools, well below the ≤ 6.83 % that applies to the held eluate. This agrees with the expectation that Protein A capture does not separate aggregate from monomer and that the short time at elution pH in the column forms little.

Residual DNA was measured on the load and the pool of the 7 centre-point runs of the two designs. The load carried 137,045 to 234,778 pg per mg antibody and the pool 498 to 1,372 pg/mg, a reduction of 2.2 to 2.5 log10 across the step. DNA was not measured away from the set-point, so its dependence on the parameters is not tested here, as PCP-005 planned.

5.5 Model residuals and influential runs

Table 5.9 gives the residual checks of criterion 5 of PCP-005 §7.

Table 5.9: Residual checks of the response-surface models (PCP-005 §7, criterion 5).

Response	p, abs. residual on predicted	p, residual on run order	Largest Cook's distance	Run	Runs above 4/n
Pool HCP (ng/mg)	0.615	0.499	0.42	1	8
Step yield	0.562	0.310	0.38	12	6
Leached Protein A (ppm)	0.789	0.340	0.48	10	2
Fragments (LMW, %, nrCE-SDS)	0.579	0.594	0.56	6	7

No model shows a slope of the absolute residuals on the predicted values (the smallest p is 0.562). No model shows a slope of the residuals on the run order of execution either (the smallest p is 0.310), so no drift over the course of the design is seen. Every model, however, has runs above the Cook's distance limit of 4/n, between 2 and 8 per response. Criterion 5 is therefore not met for any response. A full quadratic model of 15 terms in 28 runs has a mean leverage of 0.54, and the factorial and axial runs of a face-centred design carry high leverage by construction. Cook's distances above 4/n are therefore expected in this design, and the limit does not separate an influential run from a run at a corner.

For pool host cell protein, which is the only model used to set a range, each run above the limit was deleted in turn and the model refitted. The coefficients of protein load and elution buffer pH changed by at most 4 %. The highest prediction over the NOR moved between 13,132 and 13,837 ng/mg, and it stayed below the criterion of 16,000 ng/mg in every case. No single run therefore decides the conclusion for pool host cell protein (finding F7). PCP-005 provides no alternative to the Cook's distance limit, so resting the conclusion on this influence analysis is a departure from the plan, and it is listed in Section 13.

5.6 Univariate assessment

Table 5.10 gives the results of study S-PA-UNI. Each end of each range is the mean of 2 runs, and the difference is taken from the mean of the 3 set-point runs of S-PA-SDM, with its 95 % confidence interval (Student t, pooled variance).

Table 5.10: Univariate study of operating temperature and bed height (S-PA-UNI). Δ, difference from the set-point runs of S-PA-SDM, with 95 % confidence interval.

Parameter	Setting	n	HCP (ng/mg)	Δ HCP (95 % CI)	Yield (%)	Δ yield (95 % CI)	Leached PrA (ppm)	Δ leached PrA (95 % CI)
Operating temperature	15 °C	2	9,419	2,005 (-281 to 4,291)	97.1	3.6 (-4.9 to 12.0)	2.37	0.26 (-1.46 to 1.99)
Operating temperature	30 °C	2	7,831	417 (-1,186 to 2,021)	97.6	4.1 (-4.4 to 12.5)	3.62	1.51 (-0.67 to 3.70)
Bed height	10 cm	2	7,427	13 (-1,446 to 1,471)	96.7	3.1 (-6.9 to 13.0)	2.87	0.76 (-1.12 to 2.65)
Bed height	30 cm	2	7,900	486 (-1,843 to 2,815)	96.0	2.4 (-7.8 to 12.6)	2.69	0.59 (-1.15 to 2.32)

Table 5.11 gives the individual results.

Table 5.11: Individual results of study S-PA-UNI.

Parameter	Setting	Run	HCP (ng/mg)	Yield (%)	Leached PrA (ppm)
Operating temperature	15 °C	1	10,209	96.5	2.51
Operating temperature	15 °C	2	8,628	97.8	2.22
Operating temperature	30 °C	1	7,441	98.4	3.03
Operating temperature	30 °C	2	8,222	96.9	4.20
Bed height	10 cm	1	7,698	94.4	2.52
Bed height	10 cm	2	7,155	99.0	3.21
Bed height	30 cm	1	8,712	93.5	2.84
Bed height	30 cm	2	7,088	98.5	2.54

Table 5.12: Hold-time stability of the Protein A eluate (held at elution pH).

Storage	Day	Size variants (%)			Acidic variants (%)
		HMW (SEC)	LMW (SEC)	LMW (nrCE-SDS)	
2-8 °C	0	1.61	1.37[a)]	0.86	21.7
	1	1.64	1.37[a)]	n.t.[b)]	21.6
	3	1.64	1.40[a)]	n.t.[b)]	21.6
	5	1.62	1.44[a)]	n.t.[b)]	21.8
	7	1.68	1.36[a)]	1.08	22.1
21 °C	0	1.53	1.40[a)]	0.76	22.1
	1	1.71	1.43[a)]	n.t.[b)]	22.1
	3	1.68	1.41[a)]	n.t.[b)]	22.1
	5	1.86	1.49[a)]	n.t.[b)]	22.3
	7	1.98	1.61[a)]	1.08	23.0

[a)] Includes host cell protein eluting in the low molecular weight window (AMD-001). Not used for evaluation. [b)] Not tested. The first and last time points were retained and tested by nrCE-SDS after AMD-001, with the retained response-surface samples.

Every individual result met its criterion. The highest pool host cell protein was 10,209 ng/mg against ≤ 16,000 ng/mg, and the highest leached Protein A was 4.20 ppm against ≤ 158 ppm. The range studied of each parameter is therefore supported with the other parameters at their set-points.

Every confidence interval includes zero. The largest difference in pool host cell protein, 2,005 ng/mg at 15 °C of operating temperature, is not significant. With two runs at each end the intervals are wide, so the study excludes only effects larger than their upper bounds. Step yield was 2.4 to 4.1 percentage points higher than the reference at every end. Because the difference has the same sign at both ends of both parameters, it is more likely a difference between the reference runs and the univariate runs than an effect of either parameter. Neither study can detect an interaction, and a short bed at high load was not run.

5.7 Hold study of the Protein A eluate

Table 5.12 gives the results of the hold study.

The two temperatures started from aliquots of one pool, and on day 0 these already differed by 0.08 % in aggregate, 0.4 % in acidic variants and 0.10 % in fragments by nrCE-SDS (Table 5.12). This is the variation of the methods and of the aliquots, and a change smaller than it is not resolved.

At 2-8 °C the eluate changed little over 7 days. Aggregate went from 1.61 to 1.68 % and acidic variants from 21.7 to 22.1 %. At 21 °C aggregate and acidic variants rose more. Aggregate went from 1.53 to 1.98 % and acidic variants from 22.1 to 23.0 %. Fragments by nrCE-SDS rose from 0.86 to 1.08 % at 2-8 °C and from 0.76 to 1.08 % at 21 °C. Aggregate rose by 0.45 % at 21 °C against 0.07 % at 2-8 °C, which is consistent with partial unfolding at low pH that proceeds faster when warm. Fragments rose at both temperatures, by 0.22 and 0.32 %. The difference between these rises, 0.10 %,

is equal to the difference between the day 0 aliquots, so the data do not show that fragmentation depends on temperature.

Every sample met its criterion. The highest aggregate was 1.98 % against ≤ 6.83 %, acidic variants stayed within 21.6 to 23.0 % against 18–39 %, and the highest fragment result by nrCE-SDS was 1.08 % against ≤ 2.45 %. A hold of 7 days at elution pH is therefore supported at both 2-8 °C and 21 °C (finding F8). The intermediate time points have SEC fragment results only, and these were not used for the decision.

5.8 Mechanistic interpretation

Pool host cell protein rises as elution buffer pH falls, because a lower pH releases more of the host cell protein held weakly on the resin and on the antibody together with the product. A higher protein load puts more antibody on the bed, and with it more of the host cell protein associated with the antibody, so pool host cell protein rises with load. The negative AB interaction follows from the two effects acting on the same weakly bound impurity. At high load there is more of it on the bed, and a low elution pH releases more of it. The positive B^2 term shows that this release accelerates at the low end of the pH range.

End of pool collect lowers pool host cell protein per mass of antibody in the response-surface design. Collecting further into the tail adds antibody and little host cell protein, so the ratio falls. The same extension raises yield, which is the largest single effect on yield in both designs. The opposite sign of this effect in the screening design is not explained by the mechanism (finding F10).

A higher load flow rate shortens the residence time and lowers the dynamic binding capacity. This costs yield only when the bed is close to capacity, which is why the cost appears at high load and nearly vanishes at low load (the AC interaction). The mechanism also predicts a small rise in pool host cell protein at high load and high flow. The data show no such rise, and the point estimates point the other way.

Leached Protein A could not be modelled, and the mechanism gives no reason for it to follow the settings of a single run. The scale effect found in the qualification (Section 3.1) is expected to have a different cause, the longer hold of the load before capture at manufacturing scale, which this study does not test. Fragments by nrCE-SDS fall slightly as elution pH rises, which is the direction expected from acid-catalysed hinge hydrolysis during elution. The effect is small because the antibody spends only the duration of the elution at low pH in the column.

For operation, elution buffer pH and protein load have the largest effects on pool host cell protein, and the risk lies in the corner of low pH and high load. End of pool collect is also associated with pool host cell protein, but within its NOR it does not move the boundary of that corner far (Section 6). Load flow rate is associated with lower yield at high load.

6 Design space

The design space of this step is the region in protein load and elution buffer pH, the two WC-CPPs, over which pool host cell protein meets its intermediate acceptance criterion. Leached Protein A and fragments met their criteria in every run of the designs, and step yield has no criterion, so they do not constrain it. On the mean predictions of the response-surface model, with load flow rate and end of pool collect at their set-points, 96 % of the plane of protein load and elution buffer pH over the range studied meets the criterion. With load flow rate and end of pool collect at the corners of their NORs, the fraction lies between 92 and 98 %. The excluded region is the corner of high protein load and low elution buffer pH shown in Figure 5.1. At the boundary of a region defined on mean predictions, a single batch meets the criterion only about half the time (CMC Biotech Working Group 2009), so the claim is made through the PARs of Section 7, which use the 95 % prediction interval of a single run.

The ranges are nested. The NOR lies inside the characterized region, and the characterized region is the knowledge space within which a PAR can be claimed. Over the NOR the model predicts a mean pool host cell protein between 3,550 to 13,553 ng/mg (Table 7.1). The upper limit of the 95 % prediction interval of a single run reaches at most 15,621 ng/mg over the NOR, below the criterion. Movement outside the NOR and within a PAR would be assessed under change control. These statements rest on a model of the scale-down column, which is qualified for pool host cell protein (finding F1).

7 Proven acceptable ranges

7.1 Acceptance basis

Each output of the step is judged against the intermediate acceptance criterion that PCMP-001 §7.3 derives a priori. The drug substance criterion is carried backward through the train using the worst case that each downstream step showed on the platform with X-Mab, Y-Mab and Z-Mab, and no margin is applied. For pool host cell protein the drug substance limit of ≤ 100 ng/mg is carried back through the platform worst cases of anion exchange and cation exchange, which together remove host cell protein 160-fold. That gives ≤ 16,000 ng/mg. For leached Protein A the limit of ≤ 5 ppm is carried back through the same two steps, which together remove it 31.6-fold, and gives ≤ 158 ppm. For fragments the limit of ≤ 2.00 % is carried back through the lowest platform clearance of cation exchange, 1.3-fold, and the largest increase of low-pH viral inactivation, 0.15 percentage points. That gives ≤ 2.45 %. The basis of each criterion is recorded as back-calculated. PCMP-001 Table 7.1 gives the derivation for the whole train.

These criteria rest on 6 platform worst cases of the downstream steps. Every one that could be checked against a response of the downstream designs was confirmed in PCR-006, PCR-007 and PCR-008. The other 2 concern leached Protein A, which is not a response of the cation exchange or anion exchange designs, so the platform values stand. No criterion of this step was tightened. No criterion upstream of this step relies on a worst case of the Protein A step, so this report carries no verification of its own.

The decision statistic for a quality attribute is the 95 % prediction interval of a single run, because a single batch is what must meet a quality criterion (PCMP-001 §6.4). In the NOR-propagated analysis the model coefficients are drawn from their estimated covariance in every simulated run, together with the residual.

7.2 Outputs against their criteria over the NOR

Table 7.1 gives each output at the set-point and over the NOR, as predicted by the response-surface models.

All three outputs meet their criteria at the set-point and across the NOR. As the footnote to Table 7.1 states, each is judged at every combination of NOR edges and set-point on the 95 % prediction interval of a single run, which includes the uncertainty of the model coefficients and of the residual (PCMP-001 §6.4). For leached Protein A and fragments the predictions come from models that are not adequate (Section 5.3).

Table 7.1: Protein A outputs against their intermediate acceptance criteria.

Attribute	Criterion	Mean predicted by the response-surface model		Met across the NOR[a)]
		At the set-point	Range over the NOR	
Host cell protein (ng/mg)	≤ 16,000	7,216	3,550 to 13,553	Yes
Leached Protein A (ppm)	≤ 158	2.87	2.46 to 3.22	Yes
Fragments, LMW (%)	≤ 2.45	0.91	0.86 to 0.97	Yes

[a)] Judged at every combination of NOR low, set-point and NOR high. Quality attributes: the 95 % prediction interval of a single run (t quantile, coefficient and residual uncertainty). Viral clearance: the mean prediction.

For these two outputs the decision rests on the measured results, every one of which met its criterion, and the predictions are shown for completeness.

7.3 Proven acceptable ranges per attribute and parameter

Table 7.2 gives the PAR of each multivariate parameter for each output with a criterion, by the set-point analysis and by the NOR-propagated analysis of PCMP-001 §6.4. The NOR-propagated range is the one that supports a claim.

Table 7.2: Proven acceptable ranges for each output and multivariate parameter. PAR (set-point), other factors at set-point, mean prediction. PAR (NOR), other factors varied within their NOR, 95 % prediction interval of a single run.

CQA	Parameter	Char. range	Unit	Criterion	PAR (set-point)	PAR (NOR)
Pool HCP (ng/mg)	Protein load	10–50	g/L resin	≤ 16,000	10–50	10–50
Pool HCP (ng/mg)	Elution buffer pH	3.2–3.9	pH	≤ 16,000	3.2–3.9	3.26–3.9
Pool HCP (ng/mg)	Load flow rate	100–300	cm/hr	≤ 16,000	100–300	100–300
Pool HCP (ng/mg)	End of pool collect	2–3.2	CV	≤ 16,000	2–3.2	2–3.2
Leached Protein A (ppm)	Protein load	10–50	g/L resin	≤ 158	10–50	10–50
Leached Protein A (ppm)	Elution buffer pH	3.2–3.9	pH	≤ 158	3.2–3.9	3.2–3.9
Leached Protein A (ppm)	Load flow rate	100–300	cm/hr	≤ 158	100–300	100–300
Leached Protein A (ppm)	End of pool collect	2–3.2	CV	≤ 158	2–3.2	2–3.2
Fragments (LMW, %, nrCE-SDS)	Protein load	10–50	g/L resin	≤ 2.45	10–50	10–50
Fragments (LMW, %, nrCE-SDS)	Elution buffer pH	3.2–3.9	pH	≤ 2.45	3.2–3.9	3.2–3.9

CQA	Parameter	Char. range	Unit	Criterion	PAR (set-point)	PAR (NOR)
Fragments (LMW, %, nrCE-SDS)	Load flow rate	100–300	cm/hr	≤ 2.45	100–300	100–300
Fragments (LMW, %, nrCE-SDS)	End of pool collect	2–3.2	CV	≤ 2.45	2–3.2	2–3.2

For pool host cell protein the NOR-propagated PAR reaches the edge of the range studied for protein load, load flow rate and end of pool collect, and no claim is made beyond it. For elution buffer pH it is 3.26–3.9, against 3.2–3.9 in the set-point analysis. The lowest elution pH of the range fails when the other parameters move within their NOR, because a single run at high load and low pH can then exceed the criterion. The PAR of elution buffer pH still contains its NOR of 3.4–3.7 with room on both sides. The lack of fit test for this model was close to its limit (Section 5.3).

For leached Protein A and fragments the table rows come from models that are not adequate, and no PAR is claimed from them. Every run of the designs met these criteria at every corner of the range studied, so neither output constrains any parameter within it.

Figure 7.1 shows the PAR of elution buffer pH for pool host cell protein, the parameter with the largest main effect on it. The left panel shows the set-point analysis and the right panel the NOR-propagated analysis, with the acceptable region shaded green. No PAR figure is shown for leached Protein A or fragments, because no adequate model results for them.

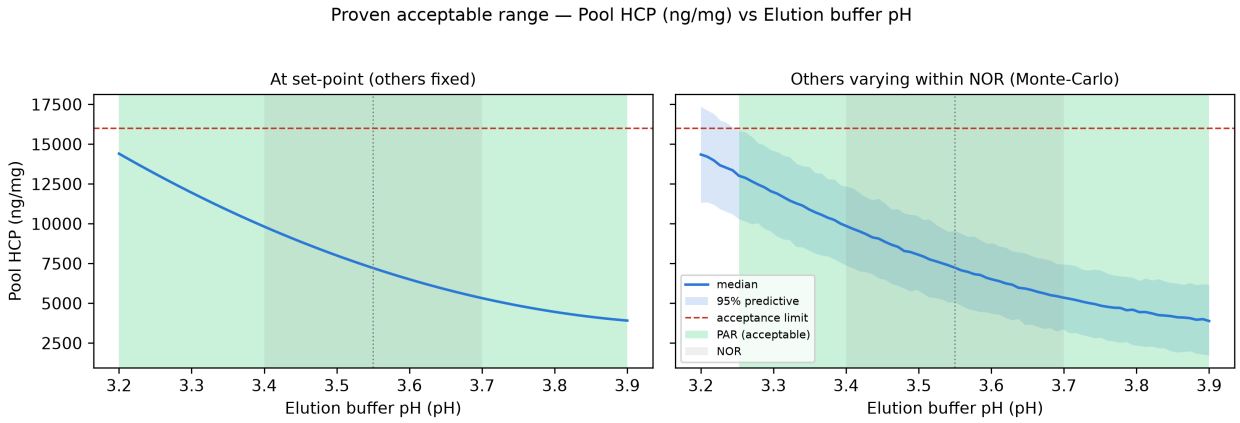


Figure 7.1: PAR of elution buffer pH for pool HCP (ng/mg). Left, set-point analysis. Right, NOR-propagated analysis (95 % prediction interval of a single run). Green, the region that meets the criterion.

Each PAR holds while the other parameters stay within their NOR, and the design space of Section 6 describes the joint region. The pool host cell protein PARs come from a scale-down model that is qualified for that output (finding F1), and they are claimed. Operating temperature and bed height have no model-based PAR. Their

ranges rest on the measured responses at each end of the range studied, with the other parameters at their set-points (Section 5.6).

8 Process capability and robustness

Capability at commercial scale was estimated from the integrated model of the train, with 2,000 simulated batches and each parameter drawn within its NOR. Table 8.1 gives the drug substance capability of the attributes to which this step contributes.

Table 8.1: Commercial-scale capability of the drug substance attributes to which the Protein A step contributes.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Host Cell Protein (HCP)	M-H	upper	≤ 100	15.2 ± 4.6	6.1
Leached Protein A	L-M	upper	≤ 5	0.003 ± 0.0006	2,785.3
Fragments (LMW, clipped species)	L	upper	≤ 2	0.634 ± 0.042	10.8

Host cell protein has the lowest Cpk of the three, 6.14. PCMP-001 §7.9 expects a Cpk of at least 1.33 of a CQA in routine manufacture, following common industry practice, and requires an action in the control strategy below 1.00. The target is met for host cell protein and fragments. Fragments reach 10.76. The Cpk of 2,785 for leached Protein A is not meaningful. The simulated drug substance values average 0.003 ppm, below the limit of quantitation of the ELISA (0.25 ppm, AMV-3016), so they cannot be measured and their spread is that of the model, not of a result. No simulated batch fell outside these criteria. These figures describe the drug substance after cation exchange and anion exchange have removed most of what the Protein A pool carries. The lowest Cpk of the drug substance, 1.03 for Acidic charge variants (deamidation), belongs to an attribute this step does not change and is discussed in PCMR-001.

Every output met its criterion across the NOR. Load flow rate is associated with lower step yield. End of pool collect is associated with higher step yield and lower pool host cell protein, and within its NOR the design space changes by a few percent of the plane (Section 6).

9 Parameter classification

Parameters were classified under SOP-4001 by the decision logic of PCMP-001 §7.7. Criterion 8 of PCP-005 §7 adds that an effect is practically relevant for an attribute when the NOR-propagated PAR of the parameter is narrower than its range studied, and that a significant effect whose PAR spans the whole range studied does not by itself make the parameter critical. The classes and the reason for each are the following.

- Protein load was classified as a WC-CPP. It raises pool host cell protein (3,872 ng/mg per half range) and interacts with elution pH. It is set by the load volume and the titre, and it is readily controlled within its NOR of 20–40 g/L resin.
- Elution buffer pH was classified as a WC-CPP. It has the largest effect on pool host cell protein (-5,245 ng/mg per half range) and the only one whose NOR-propagated PAR is narrower than its range studied. Buffer pH is controlled at preparation and verified at release. DEV-005-01 shows that a buffer below the NOR can pass release when the pH meter is incorrectly calibrated.
- Load flow rate was classified as a KPP. It lowers step yield at high load (the AC interaction) and has no significant effect on pool host cell protein, leached Protein A or fragments.
- End of pool collect was classified as a KPP. It has the largest effect on step yield. It also has a significant effect on pool host cell protein, -2,111 ng/mg across the range studied, and the sign of that effect differs between the designs (finding F10). Its NOR-propagated PAR for pool host cell protein spans the whole range studied. Within its NOR, the design space changes from 94 % of the plane at the low edge to 98 % at the high edge (Section 6). The class would be revisited if a later study confirmed the screening sign.
- Operating temperature was classified as a GPP. No difference from the set-point runs was detected at either end of its range. The intervals allow a rise in pool host cell protein of up to 4,291 ng/mg, which would still leave the highest result below the criterion, and every individual result met its criterion (Section 5.6).
- Bed height was classified as a GPP. No difference from the set-point runs was detected at either end of its range. The intervals allow a rise in pool host cell protein of up to 2,815 ng/mg, and every individual result met its criterion (Section 5.6). It is fixed by the packing specification of the commercial column and does not vary in operation. A short bed at the upper end of the load range was not characterized, and no claim is made for it.

In summary the step has 2 WC-CPPs, 2 KPPs and 2 GPPs, and no CPP. The classes of protein load and elution buffer pH rest on the direction and size of their effects, which agree with the platform history and the mechanism. The PAR of elution buffer pH for pool host cell protein is narrower than its range studied, and protein load and

elution buffer pH together form the boundary of the design space. The KPP classes rest on the direction of the yield effects in both designs and on the mechanism of breakthrough and of peak collection. They do not rest on the step yield model, which is not used under criterion 1 because the scale-down model was not shown equivalent for step yield (finding F1).

10 Contribution to the control strategy

The Protein A step contributes to the control strategy in the following ways.

1. Process parameters. Protein load and elution buffer pH are controlled within their NORs as WC-CPPs. Load flow rate and end of pool collect are controlled within their NORs as KPPs. Movement outside the NOR and within the PARs of Section 7 is assessed under change control.
2. Material controls. Elution buffer is released against its pH specification. DEV-005-01 showed that this release test depends on a correctly calibrated pH meter.
3. Resin and load. Leached Protein A depends on the resin and is expected to depend on protease activity in the load, which rises with the hold of the clarified harvest before capture (Section 3.1). Resin use, cleaning and sanitization are controlled by SOP-2008, and the hold of the clarified harvest is characterized in PCR-004.
4. Hold. The Protein A eluate may be held at elution pH for up to 7 days at 2-8 °C or 21 °C before viral inactivation.
5. Downstream clearance and release. Host cell protein, leached Protein A and fragments are further cleared by cation exchange (PCR-007) and anion exchange (PCR-008), and they are tested at release of the drug substance. Viral clearance is claimed at low-pH viral inactivation (PCR-006), anion exchange and virus filtration (PCR-009), and not at this step.

The control strategy of the whole train is consolidated in PCMR-001.

11 Discussion

The characterization confirms the main effects that the platform history predicted for this step. Elution buffer pH and protein load have the largest effects on pool host cell protein, and they interact. End of pool collect, protein load and load flow rate have the largest effects on step yield. Two results do not follow the expectations. End of pool collect lowers pool host cell protein in the response-surface design and raised it, not significantly, in the screening design, for a reason that was not identified. Load flow rate did not raise pool host cell protein as the mechanism predicts.

Confidence in the scale-down model is the weakest point of the study. For leached Protein A the qualification found a consistent difference, which is expected from the longer hold of the manufacturing load before capture and which was accepted. Even scaled by that difference the design results stay far below the criterion. The planned qualification could not demonstrate equivalence for pool host cell protein, step yield and leached Protein A, because 3 scale-down runs could not narrow the interval below the margin (DEV-005-03). A further 6 runs were made, and the margins were kept. With them the model is qualified for pool host cell protein, which is the output that sets the PARs of the step. Step yield misses equivalence by 1.4 % of its margin and column efficiency by 35 %, both on the low side. Neither sets a range, and the step yield model is not used for a range or a prediction.

The response-surface model for pool host cell protein meets criteria 2 to 4 of PCP-005 §7, with a lack of fit test close to its limit. It does not meet criterion 5, and no model does. The conclusions on pool host cell protein rest instead on the deletion of single runs, which showed that no run decides them. The Cook’s distance limit of 4/n is better suited to a design with low leverage than to a face-centred design with 15 terms in 28 runs. A limit based on the leverage of each run would serve the purpose of the criterion better. The residuals show no trend with the run order of execution (Table 5.9).

Leached Protein A and fragments gave no adequate model. For both, any effect of the parameters is small against the variation of the runs and the assay. Both met their criteria in every run, so the absence of a model limits what is known about their shape and not whether the step meets them.

The three deviations of the study did not change its conclusions (Section 13). The elution buffer of DEV-005-01 was below the NOR but inside the range studied, the leached Protein A result of DEV-005-02 was replaced by a valid re-assay, and DEV-005-03 was resolved by supplementary runs.

The other limitations are those the plan named. Each design used one feed, so the effect of feed quality on the step was not estimated. The univariate studies cannot detect interactions, and a short bed at high load was not characterized. The laboratory

column does not reproduce the hold of the manufacturing load, which raises leached Protein A at scale.

12 Conclusions

1. The step was characterized under PCP-005 in a screening design, a response-surface design, a univariate study and a hold study.
2. Elution buffer pH and protein load have the largest effects on pool host cell protein, and it falls as the end of pool collect is extended. Its model meets criteria 2 to 4 of PCP-005 §7, and its conclusions rest on an influence analysis for criterion 5. End of pool collect, protein load and load flow rate have the largest effects on step yield. Leached Protein A and fragments met their criteria in every run, and no adequate model results for either.
3. Every output met its intermediate acceptance criterion across the NOR. The NOR-propagated PAR for pool host cell protein spans the range studied for 3 parameters and is 3.26–3.9 for elution buffer pH.
4. With the supplementary runs of DEV-005-03 the scale-down model is qualified for pool host cell protein and peak asymmetry, with an accepted scale effect for leached Protein A (finding F1). Step yield and column efficiency were not shown equivalent, and neither sets a range.
5. Protein load and elution buffer pH are WC-CPPs, load flow rate and end of pool collect are KPPs, and operating temperature and bed height are GPPs. The step has no CPP.
6. No difference from the set-point runs was detected for operating temperature or bed height, and their ranges studied are supported with the other parameters at their set-points.
7. The eluate may be held at elution pH for 7 days at 2-8 °C or 21 °C.
8. At commercial scale the lowest Cpk among the attributes of this step is 6.14, for host cell protein, against the target of 1.33 set in PCMP-001.
9. The three deviations were resolved without changing the conclusions. The results roll up into PCMR-001.

13 Deviations from the plan

The register lists 3 deviations recorded for the studies of the step. All were investigated. Two were retained, and one was resolved by supplementary runs.

Deviation	Summary	Detected during	Disposition
DEV-005-01	Elution-buffer pH prepared below target on an RSM run	post-run review of the buffer preparation record (the buffer had passed release on an incorrectly calibrated pH meter)	retained
DEV-005-03	Scale-down equivalence not demonstrable with the planned three set-point runs; six supplementary runs made	Version 2.0 re-analysis of the scale-down qualification	supplementary runs executed
DEV-005-02	Leached Protein A result of RSM run 11 initially out of trend; re-assayed	analytical review	retained

Beside the recorded deviations, the analysis used two options that PCP-005 provides. The full quadratic model for step yield was replaced by a reduced model (PCP-005 §7, Section 5.3), and leached Protein A and fragments were reported descriptively (PCP-005 §5.5). The analysis departed from the plan in one place. Criterion 5 of PCP-005 §7 was not met by any model, and the conclusions on pool host cell protein rest on an influence analysis that the plan does not provide for (Section 5.5). No run was lost or invalidated, and no design was repaired or re-executed.

13.1 DEV-005-01. Elution buffer pH below target on a response-surface run

The elution buffer of lot LOT-BUF-5290 was prepared below its target pH and was used on run 23 of the response-surface design, whose design setting for elution buffer pH was 3.55. The buffer had passed release on an incorrectly calibrated pH meter. The error was found at a review of the buffer preparation record after the run, and the pH of the lot was 3.38. This is below the NOR of 3.4–3.7 and inside the range studied of 3.2–3.9. The investigation found an error at the titration end point during buffer preparation.

At the design settings of run 23 the model predicts a pool host cell protein of 8,601 ng/mg. At the measured pH it predicts 11,961 ng/mg. The run gave 8,167 ng/mg,

below the criterion of ≤ 16,000 ng/mg and closer to the prediction at the design setting than to that at the measured pH. The result therefore does not show the rise in pool host cell protein that the lower pH would cause, and the run does not distort the model in that direction. The run was analysed at its design setting, so the error in pH enters the model as part of the residual of this run. Its Cook's distance for pool host cell protein is 0.03, below the limit of 4/n. The run does not influence the fit more than the other runs of the design (Section 5.5). The run was retained. The release test is the control on which the WC-CPP class of elution buffer pH relies, and here it failed because the meter was incorrectly calibrated. Run 23 carries a footnote in Appendix B.

13.2 DEV-005-03. Scale-down equivalence not demonstrable with the planned set-point runs

PCP-005 Version 2.0 sets an equivalence margin for each output of the scale-down model qualification and plans 3 set-point runs in the scale-down model. The Version 2.0 re-analysis of the qualification found that, with 3 runs, the half-width of the 90 % confidence interval alone was larger than the margin for pool host cell protein, step yield and leached Protein A. Equivalence could not have been shown for these outputs whatever the means were. Table 13.2 gives the result on the planned runs alone, where 1 of the 5 outputs met the criterion. The root cause is a number of replicates too small for the margin the plan set.

Table 13.2: Scale-down model qualification on the planned set-point runs of S-PA-SDM alone. Abbreviations as in the qualification table of the methods section.

Output	n (SD / M)	Mean, SD	Mean, M	Difference	90 % CI	Margin	Equivalent
Pool HCP (ng/mg)	3 / 5	7,414	7,964	-550	-1,799 to 699	± 1,026	No
Step yield (%)	3 / 5	93.59	94.42	-0.83	-4.53 to 2.87	± 2.11	No
Leached Protein A (ppm)	3 / 5	2.10	3.70	-1.60	-2.76 to -0.43	± 1.08	No
Column efficiency (plates/m)	3 / 5	2,460	2,649	-188	-390 to 13	± 237	No
Peak asymmetry	3 / 5	1.22	1.22	0.00	-0.11 to 0.11	± 0.12	Yes

In study S-PA-SDM2 (2026-09-07 to 2026-09-11), 6 supplementary set-point runs were made during INV-001. The margins were not changed, because a criterion changed after the data are seen is the failure that INV-001 investigated. The equivalence test was repeated over all 9 scale-down runs (Section 3.1). The model is now qualified for

pool host cell protein and peak asymmetry. Leached Protein A differs by the accepted scale effect. Step yield and column efficiency were not shown equivalent, and neither sets a range. The supplementary runs were made after the characterization studies, so the qualification for pool host cell protein was completed after the data it supports were collected.

13.3 DEV-005-02. Leached Protein A result of run 11 initially out of trend

The leached Protein A result of run 11 of the response-surface design was out of trend at analytical review. The sample was re-assayed under AMV-3016. The investigation attributed the first result to suspected carryover in the ELISA. The re-assay result, 2.12 ppm, is the one reported in Appendix B and used in the analysis, and the first result was kept in the record with its status marked. The re-assay result lies inside the range of the other runs of the design and far below the intermediate acceptance criterion of ≤ 158 ppm.

Appendices

Appendix A. Screening design matrix and responses

Table 13.3 gives the screening design (study S-PA-SCR) in natural units, with the results.

Appendix B. Response-surface design matrix and responses

Table 13.4 gives the response-surface design (study S-PA-RSM) in natural units, with the results and the fragment results of study S-PA-RET.

Appendix C. Full effect and coefficient tables

Table 13.5 gives the screening effects on leached Protein A. The screening effects on pool host cell protein and step yield are in Table 5.4 and Table 5.5.

Table 13.5: Screening effects on leached Protein A (ppm), sorted by size.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AC	0.329	0.165	0.128	1.28	0.235	
A	-0.208	-0.104	0.128	-0.811	0.441	
BD	-0.189	-0.0944	0.128	-0.736	0.482	
BC	-0.182	-0.0912	0.128	-0.712	0.497	
C	0.18	0.0898	0.128	0.701	0.503	
D	0.152	0.0758	0.128	0.591	0.571	
AB	-0.133	-0.0664	0.128	-0.518	0.618	
CD	-0.0987	-0.0494	0.128	-0.385	0.71	
B	0.0652	0.0326	0.128	0.254	0.806	
AD	0.0567	0.0284	0.128	0.221	0.83	

Table 13.6, Table 13.7 and Table 13.8 give the full quadratic coefficients for the responses not shown in Results. The coefficients for fragments by SEC are not given, because those results are not used.

Table 13.3: Protein A screening design, settings and results.

Run	Type	Factor settings				Responses			Fragments, LMW (%)	
		Protein load (g/L resin)	Elution buffer pH (pH)	Load flow rate (cm/hr)	End of pool collect (CV)	HCP (ng/mg)	Yield (%)	Leached PrA (ppm)	SEC	nrCE-SDS
1	factorial	10	3.2	100	2	9,966	91.8	2.64	1.41	n.t.
2	factorial	10	3.2	100	3.2	9,515	99.0	3.00	1.46	n.t.
3	factorial	10	3.2	300	2	9,023	92.8	3.03	1.42	n.t.
4	factorial	10	3.2	300	3.2	7,894	98.0	2.98	1.47	n.t.
5	factorial	10	3.9	100	2	3,212	91.5	2.88	1.03	n.t.
6	factorial	10	3.9	100	3.2	2,523	98.6	3.82	1.03	n.t.
7	factorial	10	3.9	300	2	2,540	92.9	3.30	0.91	n.t.
8	factorial	10	3.9	300	3.2	3,108	96.0	2.43	0.97	n.t.
9	factorial	50	3.2	100	2	16,568	91.3	2.47	1.76	n.t.
10	factorial	50	3.2	100	3.2	19,738	99.0	2.66	1.97	n.t.
11	factorial	50	3.2	300	2	18,271	85.7	2.67	1.87	n.t.
12	factorial	50	3.2	300	3.2	21,245	93.6	3.54	1.98	n.t.
13	factorial	50	3.9	100	2	6,792	91.1	2.77	1.23	n.t.
14	factorial	50	3.9	100	3.2	7,875	97.6	2.29	1.22	n.t.
15	factorial	50	3.9	300	2	7,655	89.1	2.88	1.21	n.t.
16	factorial	50	3.9	300	3.2	7,792	92.7	3.14	1.23	n.t.
17	center	30	3.55	200	2.6	6,436	93.1	2.91	1.23	n.t.
18	center	30	3.55	200	2.6	9,652	93.3	3.07	1.36	n.t.
19	center	30	3.55	200	2.6	8,222	94.2	3.86	1.30	n.t.

LMW by SEC is biased in this matrix and is not used for evaluation (AMD-001). n.t. = not tested.

Table 13.6: Full quadratic coefficients for step yield (fraction), coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.947	0.00554	171	3.5e-23	***
A	-0.0165	0.00377	-4.36	0.000767	***
B	-0.00243	0.00377	-0.643	0.531	
C	-0.00891	0.00377	-2.36	0.0346	*
D	0.0365	0.00377	9.68	2.62e-07	***
AB	0.00266	0.004	0.665	0.517	
AC	-0.00914	0.004	-2.28	0.0399	*
AD	0.00629	0.004	1.57	0.14	
BC	0.00243	0.004	0.606	0.555	
BD	0.00291	0.004	0.727	0.48	
CD	0.00219	0.004	0.547	0.594	
A^2	-0.0128	0.00997	-1.29	0.221	
B^2	0.012	0.00997	1.2	0.252	
C^2	0.00681	0.00997	0.683	0.507	
D^2	-0.0178	0.00997	-1.78	0.0976	

Table 13.4: Protein A response-surface design, settings and results.

Run	Type	Factor settings				Responses			Fragments, LMW (%)	
		Protein load (g/L resin)	Elution buffer pH (pH)	Load flow rate (cm/hr)	End of pool collect (CV)	HCP (ng/mg)	Yield (%)	Leached PrA (ppm)	SEC	nrCE-SDS
1	factorial	10	3.2	100	2	10,011	93.0	3.42	1.41	0.90
2	factorial	10	3.2	100	3.2	9,269	99.0	3.06	1.43	1.08
3	factorial	10	3.2	300	2	10,651	92.8	2.53	1.45	1.09
4	factorial	10	3.2	300	3.2	7,715	98.8	2.48	1.36	0.94
5	factorial	10	3.9	100	2	2,993	92.3	3.27	1.05	0.93
6	factorial	10	3.9	100	3.2	2,149	97.7	1.89	0.95	0.83
7	factorial	10	3.9	300	2	2,643	90.8	4.61	1.04	0.91
8	factorial	10	3.9	300	3.2	2,663	99.0	3.13	1.07	0.93
9	factorial	50	3.2	100	2	25,114	89.0	3.38	2.28	0.90
10	factorial	50	3.2	100	3.2	20,388	99.0	2.49	1.95	1.01
11[a)]	factorial	50	3.2	300	2	22,756	86.1	2.12	2.14	1.04
12	factorial	50	3.2	300	3.2	16,694	92.4	3.03	1.84	0.90
13	factorial	50	3.9	100	2	7,882	89.5	3.63	1.24	0.77
14	factorial	50	3.9	100	3.2	6,578	97.0	2.72	1.17	0.91
15	factorial	50	3.9	300	2	6,799	84.4	3.06	1.17	0.82
16	factorial	50	3.9	300	3.2	6,100	96.2	2.82	1.20	0.89
17	axial	10	3.55	200	2.6	3,900	93.5	2.71	1.14	1.00
18	axial	50	3.55	200	2.6	9,377	93.5	3.37	1.36	0.87
19	axial	30	3.2	200	2.6	13,732	96.6	3.77	1.64	0.92
20	axial	30	3.9	200	2.6	4,122	95.5	3.80	1.08	0.85
21	axial	30	3.55	100	2.6	7,571	95.5	2.30	1.32	0.87
22	axial	30	3.55	300	2.6	8,098	95.5	2.54	1.29	0.98
23[b)]	axial	30	3.55	200	2	8,167	90.8	2.07	1.36	1.00
24	axial	30	3.55	200	3.2	6,461	95.3	2.69	1.22	0.96
25	center	30	3.55	200	2.6	8,036	94.0	2.75	1.39	0.97
26	center	30	3.55	200	2.6	7,617	96.7	1.70	1.29	0.85
27	center	30	3.55	200	2.6	7,206	92.4	4.06	1.24	0.93
28	center	30	3.55	200	2.6	7,396	95.4	2.91	1.37	0.80

[a)] DEV-005-02: Leached Protein A result of RSM run 11 initially out of trend; re-assayed. Run retained.
[b)] DEV-005-01: Elution-buffer pH prepared below target on an RSM run. Run retained.
LMW by SEC is biased in this matrix and is not used for evaluation (AMD-001). n.t. = not tested.

Table 13.7: Full quadratic coefficients for leached Protein A (ppm), coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	2.87	0.22	13	7.67e-09	***
A	-0.027	0.15	-0.18	0.86	
B	0.148	0.15	0.989	0.341	
C	0.00918	0.15	0.0612	0.952	
D	-0.21	0.15	-1.4	0.185	
AB	-0.0127	0.159	-0.0796	0.938	
AC	-0.143	0.159	-0.899	0.385	
AD	0.132	0.159	0.831	0.421	
BC	0.269	0.159	1.69	0.115	
BD	-0.226	0.159	-1.42	0.179	
CD	0.167	0.159	1.05	0.314	
A²	0.162	0.397	0.408	0.69	
B²	0.906	0.397	2.29	0.0397	*
C²	-0.458	0.397	-1.15	0.269	
D²	-0.502	0.397	-1.27	0.228	

Table 13.8: Full quadratic coefficients for fragments by nrCE-SDS (%), coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.911	0.0249	36.6	1.71e-14	***
A	-0.0275	0.017	-1.62	0.129	
B	-0.0527	0.017	-3.1	0.0084	**
C	0.0161	0.017	0.949	0.36	
D	0.00523	0.017	0.308	0.763	
AB	-0.00454	0.018	-0.252	0.805	
AC	-0.00476	0.018	-0.264	0.796	
AD	0.0146	0.018	0.81	0.433	
BC	0.00214	0.018	0.119	0.907	
BD	0.00782	0.018	0.434	0.671	
CD	-0.0321	0.018	-1.78	0.0984	
A²	0.0104	0.0449	0.233	0.82	
B²	-0.0428	0.0449	-0.954	0.357	
C²	-0.00107	0.0449	-0.0238	0.981	
D²	0.0513	0.0449	1.14	0.273	

Appendix D. Analytical methods summary

Table 13.9 summarizes the analytical methods used in the studies of the step.

Table 13.9: Analytical methods used in the studies of the Protein A step.

Technique	Analyte	AMV	Reportable	Use in this report
ELISA	Host cell protein	AMV-3012	ng per mg antibody	Response of both designs, univariate study
ELISA	Leached Protein A	AMV-3016	ppm	Response of both designs, univariate study
UV absorbance at 280 nm	Protein concentration	AMV-3019	step yield, %	Response of both designs, univariate study
Non-reduced CE-SDS	Fragments (LMW)	AMV-3020	% of total	Response of the response-surface design, hold study
SEC-HPLC	Fragments (LMW)	AMV-3011	% of total	Information only
SEC-HPLC	Aggregate (HMW)	AMV-3011	% of total	Hold study, information
icIEF	Acidic variants	AMV-3013	% of total	Hold study
qPCR	Residual DNA	AMV-3014	per mg antibody	Information only

References

CMC Biotech Working Group. 2009. “A-Mab: A Case Study in Bioprocess Development.” CASSS.

International Council for Harmonisation. 2009. “ICH Q8(R2): Pharmaceutical Development.” ICH.

———. 2012. “ICH Q11: Development and Manufacture of Drug Substances.” ICH.

———. 2023. “ICH Q9(R1): Quality Risk Management.” ICH.

U.S. Food and Drug Administration. 2011. “Guidance for Industry: Process Validation — General Principles and Practices.” FDA.